Foundations of Ridge regression

[Ridge regression]

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$$

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

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β ^ = X + Y , ς ^ 2 = ■ Y − X β ^ ■ 2 n − p . {\displaystyle {\hat {\boldsymbol {\beta }}}=\mathbf {X} ^{+}\mathbf {Y} ,\qquad {\hat {\varsigma }}^{2}={\frac {\|\mathbf {Y} -\mathbf {X} {\hat {\boldsymbol {\beta }}}\|^{2}}{n-p}}.}

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where Y {\displaystyle \mathbf {Y} } is the regressand or response vector, X {\displaystyle \mathbf {X} } is the design matrix, I {\displaystyle \mathbf {I} } is the identity matrix, and the ridge (or Tikhonov) regularization parameter λ ≥ 0 {\displaystyle \lambda \geq 0} serves as the constant shifting the diagonals of the moment matrix. It can be shown that this estimator is the solution to the least squares problem subject to the constraint β T β = c {\displaystyle {\boldsymbol {\beta }}^{\mathsf {T}}{\boldsymbol {\beta }}=c} , which can be expressed as a Lagrangian minimization:

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where A = (A Γ) {\displaystyle {\mathcal {A}}={\begin{pmatrix}A\\\Gamma \end{pmatrix}}} and b = (b 0) {\displaystyle {\mathcal {b}}={\begin{pmatrix}\mathbf {b} \\{\boldsymbol {0}}\end{pmatrix}}} , for some suitably chosen Tikhonov matrix Γ {\displaystyle \Gamma } . In many cases, this matrix is chosen as a scalar multiple of the identity matrix (Γ = α I {\displaystyle \Gamma =\alpha I}), giving preference to solutions with smaller norms; this is known as L2 regularization. In other cases, high-pass operators (e.g., a difference operator or a weighted Fourier operator) may be used to enforce smoothness if the underlying vector is believed to be mostly continuous. This regularization improves the conditioning of the problem, thus enabling a direct numerical solution. Treating it as an ordinary least squares problem with augmented matrices A {\displaystyle {\mathcal {A}}} and b {\displaystyle {\mathcal {b}}} , the solution is

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which shows that λ {\displaystyle \lambda } is nothing but the Lagrange multiplier of the constraint. In fact, there is a one-to-one relationship between c {\displaystyle c} and λ {\displaystyle \lambda } and since, in practice, we do not know c {\displaystyle c} , we define λ {\displaystyle \lambda } heuristically or find it via additional data-fitting strategies, see Determination of the Tikhonov parameter below. Note that as λ ↓ 0 {\displaystyle \lambda \downarrow 0} , the constraint eventually becomes non-binding, and the ridge estimator converges to the minimum-norm ordinary least squares estimator, here denoted as β ^ = β ^ 0 {\displaystyle {\hat {\boldsymbol {\beta }}}={\hat {\boldsymbol {\beta }}}_{0}} :

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where x {\displaystyle \mathbf {x} } and b {\displaystyle \mathbf {b} } may be of different sizes and A {\displaystyle A} may even be non-square. The standard approach is ordinary least squares linear regression. However, if no x {\displaystyle \mathbf {x} } satisfies the equation or more than one x {\displaystyle \mathbf {x} } does—that is, the solution is not unique—the problem is said to be ill posed. In such cases, ordinary least squares estimation leads to an overdetermined, or more often an underdetermined system of equations. Most real-world phenomena have the effect of low-pass filters in the forward direction where A {\displaystyle A} maps x {\displaystyle \mathbf {x} } to b {\displaystyle \mathbf {b} } . Therefore, in solving the inverse-problem, the inverse mapping operates as a high-pass filter that has the undesirable tendency of amplifying noise (eigenvalues / singular values are largest in the reverse mapping where they were smallest in the forward mapping). In addition, ordinary least squares implicitly nullifies every element of the reconstructed version of x {\displaystyle \mathbf {x} } that is in the null-space of A {\displaystyle A} , rather than allowing for a model to be used as a prior for x {\displaystyle \mathbf {x} } . Ordinary least squares seeks to minimize the sum of squared residuals, which can be compactly written as

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The Lavrentyev regularization, if applicable, is advantageous to the original Tikhonov regularization, since the Lavrentyev matrix A + Q {\displaystyle A+Q} can be better conditioned, i.e., have a smaller condition number, compared to the Tikhonov matrix A T A + Γ T Γ . {\displaystyle A^{\mathsf {T}}A+\Gamma ^{\mathsf {T}}\Gamma .}

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lim λ ↓ 0 β ^ λ = X + Y = β ^ 0 , {\displaystyle \lim _{\lambda \downarrow 0}{\hat {\boldsymbol {\beta }}}_{\lambda }=\mathbf {X} ^{+}\mathbf {Y} ={\hat {\boldsymbol {\beta }}}_{0},} with X + {\displaystyle \mathbf {X} ^{+}} denoting the pseudoinverse of X {\displaystyle \mathbf {X} } .

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x ^ = (A T A) − 1 A T b = (A T A + Γ T Γ) − 1 A T b . {\displaystyle {\hat {\mathbf {x} }}=({\mathcal {A}}^{\mathsf {T}}{\mathcal {A}})^{-1}{\mathcal {A}}^{\mathsf {T}}\mathbf {\mathcal {b}} =(A^{\mathsf {T}}A+\Gamma ^{\mathsf {T}}\Gamma)^{-1}A^{\mathsf {T}}\mathbf {b} .}

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Regularization in Hilbert space Typically discrete linear ill-conditioned problems result from discretization of integral equations, and one can formulate a Tikhonov regularization in the original infinite-dimensional context. In the above we can interpret A {\displaystyle A} as a compact operator on Hilbert spaces, and x {\displaystyle x} and b {\displaystyle b} as elements in the domain and range of A {\displaystyle A} . The operator A ∗ A + Γ T Γ {\displaystyle A^{*}A+\Gamma ^{\mathsf {T}}\Gamma } is then a self-adjoint bounded invertible operator.

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the problem of a near-singular moment matrix X T X {\displaystyle \mathbf {X} ^{\mathsf {T}}\mathbf {X} } is alleviated by adding positive elements to the diagonals, thereby decreasing its condition number. Compared to the ordinary least squares estimator, the simple ridge estimator has an extra term λ I {\displaystyle \lambda \mathbf {I} } in the denominator:

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G = RSS τ 2 = ■ X β ^ − Y ■ 2 [tr ■ (I − X (X T X + λ 2 I) − 1 X T)] 2 , {\displaystyle G={\frac {\operatorname {RSS} }{\tau ^{2}}}={\frac {\left\|\mathbf {X} {\hat {\boldsymbol {\beta }}}-\mathbf {Y} \right\|^{2}}{\left[\operatorname {tr} \left(\mathbf {I} -\mathbf {X} \left(\mathbf {X} ^{\mathsf {T}}\mathbf {X} +\lambda ^{2}\mathbf {I} \right)^{-1}\mathbf {X} ^{\mathsf {T}}\right)\right]^{2}}},}

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Generalized Tikhonov regularization For general multivariate normal distributions for x {\displaystyle \mathbf {x} } and the data error, one can apply a transformation of the variables to reduce to the case above. Equivalently, one can seek an x {\displaystyle \mathbf {x} } to minimize